Expt 3 Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

Objectives:

1. To design Butterworth and Chebyshev Type-I IIR Band-Pass Filters in MATLAB for the given passband, stopband, ripple, attenuation, and sampling frequency specifications.
2. To determine the minimum filter order and corresponding cutoff frequencies and obtain the filter coefficients for both IIR filter designs.
3. To analyze and compare the frequency-domain characteristics of the Butterworth and Chebyshev Type-I filters using magnitude response, phase response, and group delay plots.
4. To evaluate the stability and performance of the designed filters using pole-zero plots, impulse responses, and pole magnitude analysis.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the filter specifications, including sampling frequency, passband frequencies, stopband frequencies, passband ripple, and stopband attenuation.
3. Normalize the passband and stopband frequencies with respect to the Nyquist frequency (Fs/2).
4. Use the buttord() function to determine the minimum order and cutoff frequencies of the Butterworth Band-Pass Filter.
5. Design the Butterworth Band-Pass Filter using the butter() function and obtain its numerator and denominator coefficients.
6. Use the cheb1ord() function to determine the minimum order and cutoff frequencies of the Chebyshev Type-I Band-Pass Filter.
7. Design the Chebyshev Type-I Band-Pass Filter using the cheby1() function and obtain its numerator and denominator coefficients.
8. Display the filter order, cutoff frequencies, and filter coefficients for both filters.

9. Plot and analyze the frequency responses of the Butterworth and Chebyshev Type-I filters

using the freqz() function.

10. Compare the magnitude responses and phase responses of both filters using frequency response plots.

11. Generate pole-zero plots using the zplane() function and observe the pole and zero locations.

12. Obtain and compare the group delay characteristics using the grpdelay() function.

13. Plot the impulse responses of both filters using the impz() function.

14. Determine the pole magnitudes using the roots() function and verify filter stability by checking whether all poles lie inside the unit circle.

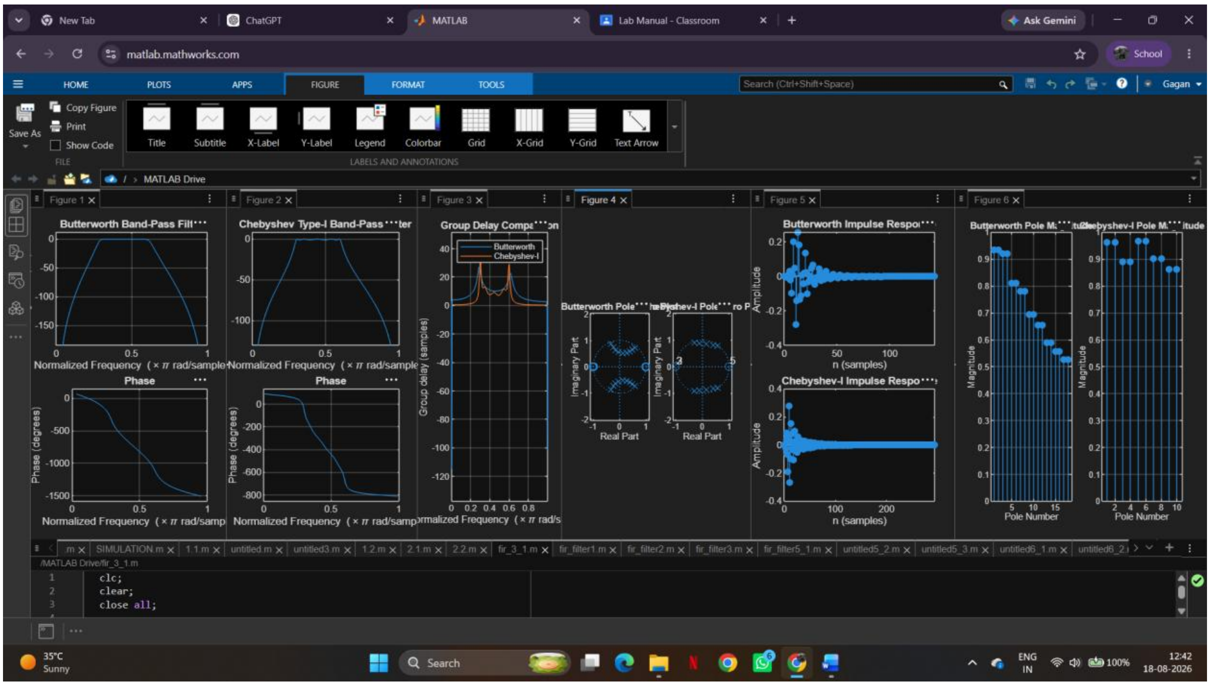
15. Analyze and compare the overall performance of the Butterworth and Chebyshev Type-I Band-Pass Filters based on their frequency responses, group delay, impulse response, and stability characteristics.

Program (Matlab):

```
clc; clear; close all;

% Specifications

Fs=2000; Rp=1; Rs=40;

Wp=[300 600]/(Fs/2);

Ws=[200 700]/(Fs/2);


% Butterworth Filter

[n1,W1]=buttord(Wp,Ws,Rp,Rs);

[b1,a1]=butter(n1,W1,'bandpass');


% Chebyshev Type-I Filter

[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);

[b2,a2]=cheby1(n2,Rp,W2,'bandpass');


% Magnitude Response

figure;

freqz(b1,a1); title('Butterworth Filter');
```

```
figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])
```

Output:



Butterworth Order = 9

Chebyshev Order = 5

Results:

1. Butterworth and Chebyshev Type-I IIR Band-Pass Filters were successfully designed and implemented in MATLAB.

2. The minimum filter orders and cutoff frequencies were obtained for both filters.

3. The frequency responses were analyzed and compared; the Chebyshev filter provided a sharper transition, while the Butterworth filter showed a smoother response.

4. Pole-zero analysis confirmed that both filters are stable and satisfy the given design specifications.